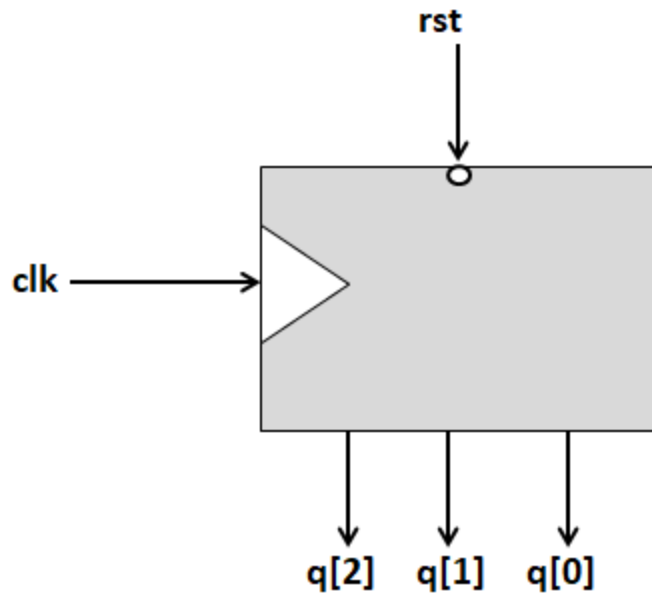


Contents of the lecture

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- (3) Verilog Code of Positive edge triggered 3 Bit Down Counter
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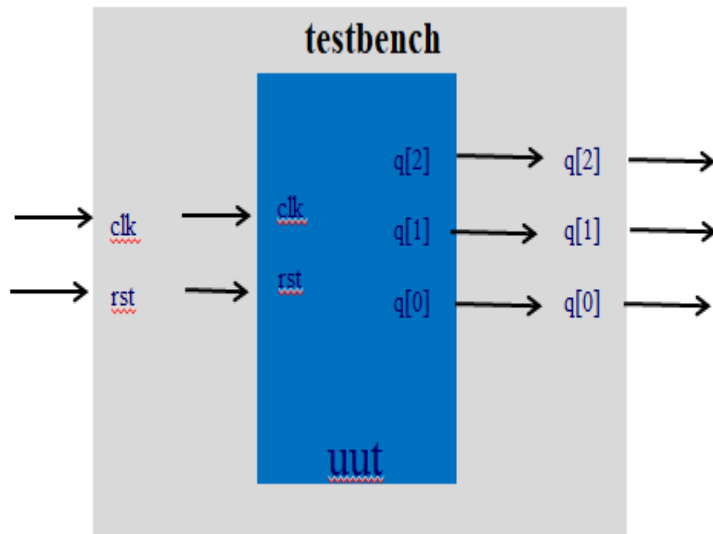


Verilog Code of Positive edge triggered 3 Bit Up Counter

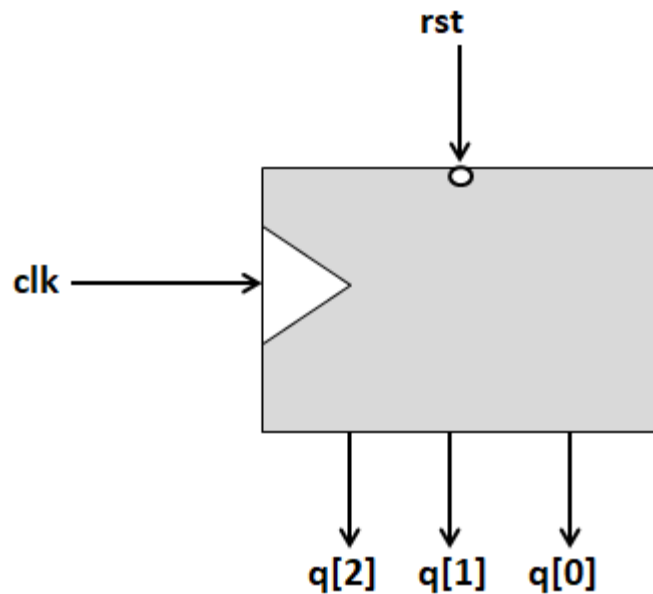


```
module up_3bit_counter(clk,rst,q);  
input clk,rst;  
output [2:0]q;  
reg [2:0]q;  
always@(negedge rst,posedge clk)  
if (!rst)  
q=3'b000;  
else  
q=q+001;  
endmodule
```

```
module up_3bit_counter(clk,rst,q);
input clk,rst;
output [2:0]q;
reg [2:0]q;
always@(negedge rst,posedge clk)
if (!rst)
q=3'b000;
else
q=q+001;
endmodule
```

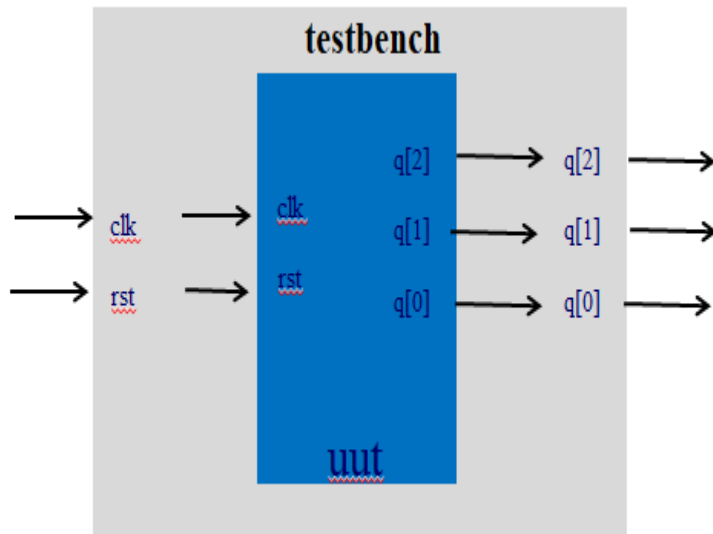


```
module testbench;
reg rst; reg clk; wire [2:0]q;
up_3bit_counter uut (.rst(rst),.clk(clk),.q(q));
initial
$monitor($time,"rst=%b,clk=%b,q=%b",rst,clk,q);
initial
begin
rst=1'b0;
repeat(40)
#40 rst= ~rst;
end
initial
begin
clk=1'b0;
repeat(100)
#2 clk= ~clk;
end
initial
#200 $finish;
endmodule
```



```
module up_3bit_counter(clk,rst,q);  
input clk,rst;  
output [2:0]q;  
reg [2:0]q;  
always@(negedge rst,posedge clk)  
if (!rst)  
q=3'b000;  
else  
q=q-001;  
endmodule
```

```
module up_3bit_counter(clk,rst,q);
input clk,rst;
output [2:0]q;
reg [2:0]q;
always@(negedge rst,posedge clk)
if (!rst)
q=3'b000;
else
q=q+001;
endmodule
```



```
module testbench;
reg rst; reg clk; wire [2:0]q;
up_3bit_counter uut (.rst(rst),.clk(clk),.q(q));
initial
$monitor($time,"rst=%b,clk=%b,q=%b",rst,clk,q);
initial
begin
rst=1'b0;
repeat(40)
#40 rst= ~rst;
end
initial
begin
clk=1'b0;
repeat(100)
#2 clk= ~clk;
end
initial
#200 $finish;
endmodule
```